Associations Among Emotional State, Sleep Quality, and Resting-State EEG Spectra: A Longitudinal Study in Graduate Students

Oleksii Komarov, Li-Wei Ko, and Tzyy-Ping Jung

***Abstract*— University students are routinely influenced by a variety of natural stressors and experience irregular sleep-wake cycles caused by the necessity to trade sleep for studying while dealing with academic assignments. Often these factors result in long-term issues with daytime sleepiness, emotional instability, and mental exhaustion, which may lead to difficulties in the educational process. This study introduces the Daily Sampling System (DSS) implemented as a smartphone application, which combines a set of self-assessment scales for evaluating variations in the emotional state and sleep quality throughout a full academic term. In addition to submitting the daily sampling scores, the participants regularly filled in the Depression, Anxiety, and Stress Scales (DASS) reports and took part in resting-state EEG data recording immediately after report completion. In total, this study collected 1835 daily samples and 94 combined DASS with EEG datasets from 18 university students (aged 23-27 years), with 79.3±15.3% response ratio in submitting the daily reports during an academic semester. The results of pairwise testing and multiple regression analysis demonstrate that the daily level of self-perceived fatigue correlates positively with stress, daytime sleepiness, and negatively with alertness on awakening, self-evaluated sleep quality, and sleep duration. The spectral analysis of the EEG data reveals a significant increase in the resting-state spectral power density across the theta and low-alpha frequency bands associated with increased levels of anxiety and stress. Additionally, the state of depression was accompanied by an intensification of high-frequency EEG activity over the temporal regions. No significant differences in prefrontal alpha power asymmetry were observed under the described experimental conditions while comparing the states of calmness and emotional arousal of the participants for the three conditions of depression, anxiety, and stress.**

***Index Terms*— Students, stress, fatigue, anxiety, depression, sleep, electroencephalography.**

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I. Introduction

STUDENTS are a high-risk group for experiencing the negative effects of stress that are often accompanied by sleep deprivation. According to a survey of 95,761 American college students in 2016 [1], more than half of the participants reported long-term problems with stress, tiredness, and sleepiness during daytime activities; 30.1% of the participants considered sleep difficulties traumatic or very difficult to handle; 43.3% rated the overall level of stress experienced within the past academic year as more than average and 11.4% of the students surveyed reported being in the state of continuous, tremendous stress. Irregular sleep-wake patterns and prolonged stress are considered as factors associated with a decline in academic performance [2], [3], deviations in eating behavior [4], [5], and mental health issues [6]–[8].

Concerns about academic performance, post-graduation plans, and competitive pressure, as well as a feeling of social insecurity, financial troubles, and expectations from their families, are among the reasons for experiencing emotional vulnerability in students [9], [10]. A prolonged cognitive load may lead to an acute mental disorder called a mental or nervous breakdown, which typically occurs in the case of severe stress-induced anxiety or depression [11] and is characterized by a failure to perform daily life activities and professional duties. It can be accompanied by insomnia and occupational burnout, which often results in dropping out of college without obtaining a degree [12], [13]. Another dangerous phenomenon is suicidal thoughts and behaviors [14]: 8.1-53.0% and

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1.0-11.2% surveyed in different studies university students reported suicidal ideation and attempts, respectively [15]–[17].

Fatigue is a commonly used term that denotes the feeling of tiredness, exhaustion, and lack of energy. Cohort studies have reported fifteen to thirty percent of participants surveyed to feel tired most of the time, with a higher prevalence of self-assessed fatigue among women [18], [19]. Despite the deterioration in prolonged task performance being referred to as fatigue in both physical and mental fatigue senses, their underlying physiological mechanisms are different. It is believed that short-term mental fatigue can be related to either a decrease over time in the availability of cognitive resources while executing a task, in a way analogous to physical fatigue, or a lack of motivation and task engagement [20], [21]. Mental fatigue is commonly associated with difficulties in maintaining a high level of sustained attention, which leads to an inability to suppress irrelevant information and prevent distractions, in particular during educational activities [22], [23].

Portable and wireless consumer-grade EEG devices have gained particular popularity over the past decade - both because of developments in the wearable sensor technologies and availability of computational power for the application of advanced signal processing algorithms [24]. As distinct changes in the EEG signal properties accompany variations in the brain dynamics, it is possible to recognize them for designing a passive or implicit brain-computer interface capable of predicting the cognitive state of its operator [25], [26]. The reported features of interest related to emotional state cover alterations in the alpha wave spectral power that is often exhibited as asymmetry in prefrontal cortex activations [27]–[29], which reflects synaptic changes in a physiological response to stress and depression [30], [31].

Currently, most of the information about relationships between psychological state, sleep, and fatigue in the general population has been obtained from survey-based research [32], [33]. This study employed longitudinal monitoring of the self-assessed fatigue, stress, and sleep quality measures in a group of graduate students during an entire academic term using the Daily Sampling System (DSS), which was implemented as a smartphone application that enabled a quick filling-in and submission of daily reports. To investigate electrophysiological correlates of the students' mental state changes over time, the resting-state EEG signals, and Depression, Anxiety, and Stress Scales (DASS) were recorded from the participants in multiple experimental sessions throughout the full duration of the study.

II. Material and Methods

A. Participants

Eighteen graduate students (24.0±1.2 years old, ten male and eight female) at National Chiao Tung University, Hsinchu, Taiwan (NCTU) were recruited to take part in this study during the first semester of the 2014/2015 academic year (start date 2014-09-15). Prior to the data collection, the participants were instructed on how to fill in and submit daily reports using the DSS application, and signed a written informed consent form. Every month each subject received a progressive monetary reward for taking part in the experiment depending on the number of submitted reports: 30 TWD (approximately equivalent to 1 USD or 0.85 EUR during the time of the experiment) per each submitted DSS report for the first 10 reports in a month, 40 TWD for each of 11-20 daily reports, and 50 TWD for each report over the 20 previously submitted reports (for example, 23 submitted reports in a month would be evaluated as 850 TWD). In addition to reporting the DSS measures, the participants were invited to complete a DASS-21 assessment and take part in resting-state EEG data collection every two weeks throughout the academic semester, during their routine classroom activities. The EEG collection sessions were scheduled for the same time and day of the week for each subject to minimize the impact of circadian rhythms. Taking part in the EEG collection routine was additionally compensated as 1000 TWD for each session. The experimental protocol was approved by the Institutional Review Board of NCTU and given the number NCTU-REC-103-025.

B. Daily Sampling System

To assess daily fluctuations in stress, fatigue, sleep quality, and daytime sleepiness in a long-term period, the participants were requested to fill in and submit, to a server, one DSS report every morning before their classes, via the smartphone application developed for this study. The application's user interface and collected DSS indices are described below.

1) Self-Evaluated Fatigue and Stress Scores: The reports were visually implemented in the application as slider bars on horizontal lines with labels 'no fatigue' to 'extreme fatigue' and 'no stress' to 'extreme stress', at the left and right ends, respectively. Each scale is from 0 (no effect present) to 100 (severe effect).

2) Daytime Sleepiness: The test for daytime sleepiness was identical to the pencil-and-paper version of Karolinska Sleepiness Scale (KSS) [34], which measures the subjective level of sleepiness at a particular time. Radio button selectors were provided next to each of the response options on a 9-point Likert-type scale, from 1 (extremely alert) to 9 (extremely sleepy, fighting sleep).

3) Sleep Quality: It was requested that participants evaluate their sleep duration and quality using a questionnaire based on the Pittsburgh Sleep Diary (PSD) [35]. The 'ratings' items from the PSD were implemented as visual scales similar to the stress and fatigue measures, indicating subjective sleep quality (on a scale of 'very bad' to 'very good'), mood on final awakening (from 'very tense' to 'very calm'), and alertness on final awakening (from 'very bad' to 'very good', superimposed onto 0 to 100 point scales. Bed and wake times were input using drop-down menus in 24-hour time format.

Each daily report was sent to the server immediately after completion. The participants could not review the reports of other participants and their own previously submitted reports. The amount of time required to complete and submit a daily report was evaluated for a single subject in a pilot experiment, and was consistently less than five minutes.

C. Depression, Anxiety, and Stress Scales

The Depression, Anxiety, and Stress Scales used in this experiment (DASS) is a standardized assessment tool that

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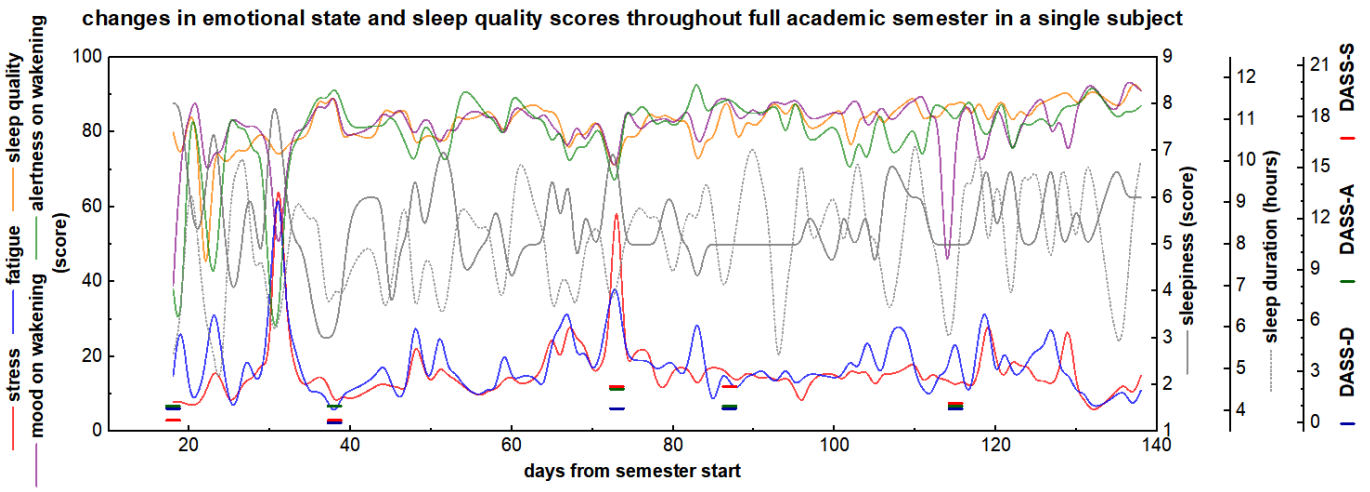

Fig. 1. Changes over time in the DSS indices, based on 112 daily reports collected during one academic semester from a single subject (subject 18, Table I). The data points in this plot are interpolated with basis splines for visual clarity. The stress, fatigue, daytime sleepiness, sleep quality, mood, and alertness scores are self-evaluated; the sleep duration is calculated as the difference between go-to-sleep and wake-up times reported for each night. The participant submitted his first daily report on the 18th day of the semester.

includes three subscales to assess the dimensional components of depression (DASS-D), anxiety (DASS-A), and stress (DASS-S) conditions [36]. It achieves a better separability between the measures of anxiety and depression in comparison with other commonly used tests, such as the Hamilton Scales for Anxiety and Depression, State-Trait Anxiety Inventory, and Beck Anxiety and Depression Inventories [37], and is considered to be an effective and easy way to administer psychometric measures applicable to both non-clinical and clinical groups [38]–[40]. The depression scale includes the items that measure symptoms typically associated with dysphoric moods, such as sadness or worthlessness. The anxiety scale includes items that are primarily related to symptoms of physical arousal, panic attacks, and fear. The stress scale evaluates such factors as tension, irritability, and tendency to overreact to stressful events. A standard, shorter 21-item version (DASS-21) of the original 42-item questionnaire was used in this study [37].

D. Electroencephalography

Five minutes of eyes-open resting-state EEG were collected from the subjects immediately after the DASS-21 completion. The signal acquisition was carried out using a Compumedics Neuroscan system, with a Scan NuAmps amplifier set at a sampling rate of 1000 Hz. Thirty sensors were placed on the head using a 32-channel Quik-Cap with averaged mastoid reference in accordance with the Modified Combinatorial Nomenclature, an extension of the International 10-20 System. The skin under the reference electrodes was abraded using Nuprep solution (Weaver and Co., United States) and cleaned with a 70% alcohol pad. The impedance between electrodes and skin was set below 5 $k\Omega$ using NaCl-based conductive gel (Quik-Gel, Neuromedical Supplies). The data processing was performed using EEGLAB 15.0b [41] installed as a toolbox for MATLAB 2018b. The collected EEG signals were inspected for noise-corrupted and fallen-off channels, subjected to 1 Hz high-pass and 50 Hz low-pass basic FIR filters, to remove the baseline drift and power grid noise, then down-sampled to 250 Hz for further processing.

E. Statistical Analysis

Differences between the group averaged values were validated using analysis of variance with an alpha level of 0.05 for a null hypothesis rejection and false discovery rate correction for multiple comparisons [42]. The outliers detection was performed using Tukey's method, with the fence constant $k = 1.5$. For multiple regression model fit the reported by each participant Daily Sampling scores were standardized using z-score normalization.

III. Results

A. Dataset Overview

In total, this study collected 1,835 daily reports and 94 combined DASS-21 with EEG datasets from 18 participants, resulting in (mean±SD) 102±20 and 5.2±1.9 reported samples per subject, respectively. The participants proceeded to take part in the data collection as they were recruited: the very first daily report in this study was submitted nine days after the semester started, while the final recruited subject filled in his first report on the eighteenth day. The response ratio was calculated individually for each participant as a ratio of the received reports to the total number of days in the experiment. Figure 1 visualizes changes over time in the indices collected for a single subject. No significant gender difference in the averaged across study duration self-assessed levels of fatigue, stress, sleep quality, mood, alertness, and sleep duration was observed in the experimental group. The data collected from all male and female subjects were merged to improve the dataset's generalizability in subsequent analysis. Table I summarizes the information collected for each participant, which includes gender, age, daily report submission ratio, number of combined DASS-21 with EEG sessions, and DSS components, averaged over the entire study duration.

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TABLE I
SUMMARY OF COLLECTED DAILY SAMPLING DATA BY SUBJECT

subject ID	gender	age	daily reports submitted	response ratio	DASS-21 / EEG sessions	daily sampling index, mean±SD						
						stress	fatigue	sleepiness	sleep quality	mood	alertness	sleep duration (h)
1	male	25	126	96.2%	6	74±7	75±10	6.7±0.8	67±15	68±15	66±14	7.7±1.6
2	male	24	79	60.3%	6	51±14	47±14	4.7±1.2	52±12	46±10	51±10	6.8±1.4
3	female	24	90	69.2%	8	47±10	49±10	5.9±0.9	51±7	50±8	50±9	6.6±1.5
4	female	24	87	66.9%	7	47±10	63±10	6.7±0.8	56±11	51±8	36±10	6.6±1.3
5	female	27	107	81.1%	7	60±18	44±18	4.3±1.6	66±14	66±14	57±20	8.0±1.8
6	female	23	113	89.7%	6	39±10	39±11	6.3±1.2	56±13	59±12	56±13	7.3±1.1
7	female	23	109	87.2%	5	36±10	33±7	6.5±1.9	74±16	56±9	49±14	7.5±1.6
8	female	26	119	90.2%	6	21±11	20±10	3.1±1.2	51±16	70±13	68±14	8.0±1.1
9	female	25	86	66.2%	6	40±20	41±20	4.3±2.0	59±15	64±12	54±18	8.3±1.6
10	male	25	104	80.0%	4	80±19	78±22	8.6±1.0	66±26	27±24	25±27	7.7±1.2
11	male	23	129	99.2%	6	62±11	60±12	7.2±0.9	60±15	42±11	41±11	6.8±1.1
12	female	23	122	94.6%	6	35±24	45±27	3.7±1.8	73±21	73±22	65±26	6.8±1.3
13	male	23	113	86.3%	6	64±8	63±8	7.5±0.9	52±9	52±9	51±7	7.2±1.5
14	male	23	94	71.8%	3	58±8	59±11	6.7±1.1	58±16	51±14	49±14	7.3±1.3
15	male	23	88	70.4%	–	38±17	36±18	8.1±1.4	56±24	56±22	50±23	7.6±2.2
16	male	23	49	39.5%	2	68±15	66±17	6.8±1.2	40±18	38±17	32±18	7.4±0.9
17	male	24	108	87.8%	5	52±22	70±16	7.2±1.0	53±17	52±20	46±19	8.0±1.7
18	male	24	112	91.1%	5	16±10	18±10	5.4±1.1	82±8	82±10	80±13	8.0±1.6

Note. Stress, fatigue, sleep quality, mood on awakening, and alertness on awakening are self-evaluated scores according to visual scales from 0 to 100. The level of daytime sleepiness is from 1 (extremely alert) to 9 (extremely sleepy); sleep duration is measured, in hours, as the difference between reported go-to-bed and wake-up times.

The macro-averaged across the group daily sampling scores are as follows: 49.4±17.4 for stress, 50.3±17.6 for fatigue, 6.1±1.5 for daytime sleepiness, 59.6±10.2 for sleep quality, 55.6±13.3 for mood on awakening, 51.4±13.2 for alertness on awakening, 7.4±0.5 hours for sleep duration.

Upon inspecting the sleep duration datasets, two common mistakes were discovered in the daily reports filling by the participants, which caused incorrect determinations of the sleep durations, inverted 'waketime' and 'bedtime' inputs and issues in dealing with 24-hour time format. The dataset was inspected for the described errors and 2.9% of all sleep duration reports were rejected from subsequent analysis because of the aforementioned reasons. All nighttime sleep duration reports collected in this study are distributed with the mean value of 7.4 hours and standard deviation of 1.5 hours; the first quartile, median and third quartile values are 6.5, 7.3, and 8.2 hours respectively. The histogram depicting the distribution of all correctly filled sleep duration reports is included in supplementary materials.

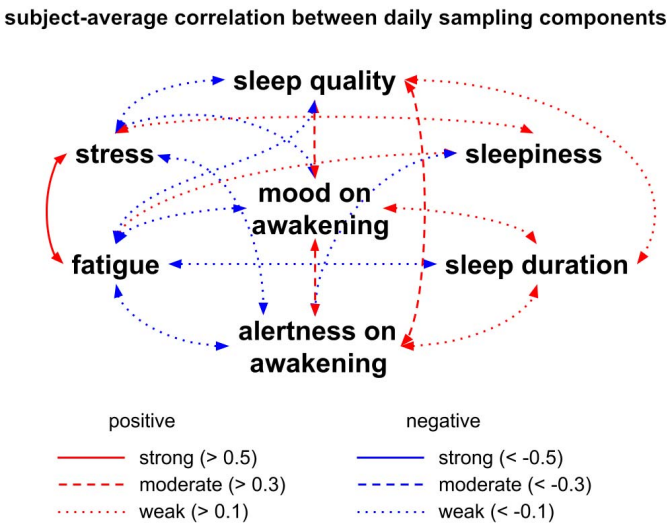

Fig. 2. Degree of bivariate correlation for all pairwise combinations of the DSS indices based upon inter-subject macro-averaging. The map is built using positive or negative correlation values of 0.1, 0.3, and 0.5 as the thresholds for weak, moderate, and strong levels respectively ($p<0.05$).

B. Bivariate Correlation Analysis

To evaluate individual variability in mutual relationships between components of the Daily Sampling System, all longitudinally collected data from individual subjects (as illustrated for a single subject in Figure 1) were used for pairwise analysis for all combinations of the six self-assessed scores and sleep duration. The results of bivariate correlation analysis are summarized in Figure 2, considering the values of 0.5, 0.3 and 0.1 as the thresholds for strong, moderate and weak degree of correlation [43]. The self-assessed daily level of stress exhibits a strong positive correlation with fatigue (0.626±0.233, positive for all subjects) and a weak negative correlation with the mood on awakening (−0.335±0.264), alertness on awakening (-0.299±0.308), and sleep quality (−0.237±0.255). The fatigue score correlates positively with daytime sleepiness (0.403±0.311), and negatively with alertness on awakening (−0.414±0.360), mood (−0.290±0.284), sleep quality (−0.267±0.233) and duration (−0.248±0.230).

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TABLE II
GROUP-AVERAGE BIVARIATE CORRELATION FOR ALL PAIRWISE COMBINATIONS OF THE DAILY SAMPLING COMPONENTS

	Fatigue	Stress	Sleepiness	Sleep quality	Mood	Alertness	Sleep duration
Fatigue	—						
Stress	0.626 ± 0.233	—					
Sleepiness	0.403 ± 0.311	0.253 ± 0.300	—				
Sleep quality	-0.267 ± 0.233	-0.237 ± 0.255	-0.146 ± 0.341	—			
Mood	-0.290 ± 0.284	-0.335 ± 0.264	-0.185 ± 0.347	0.522 ± 0.278	—		
Alertness	-0.414 ± 0.360	-0.299 ± 0.308	-0.402 ± 0.414	0.432 ± 0.251	0.564 ± 0.305	—	
Sleep duration	-0.248 ± 0.230	-0.173 ± 0.180	-0.210 ± 0.332	0.260 ± 0.174	0.316 ± 0.164	0.395 ± 0.214	—

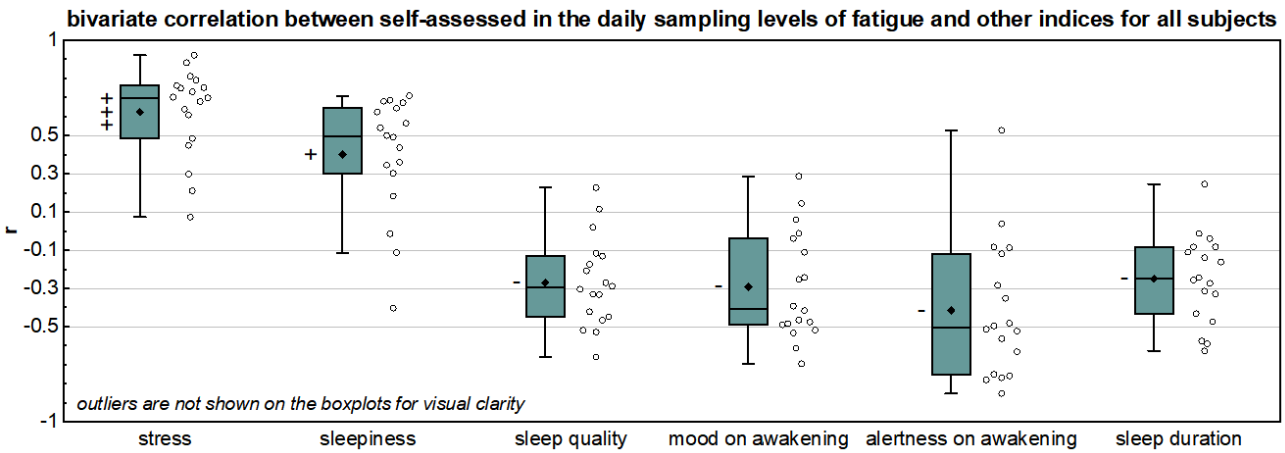

Fig. 3. Distributions of bivariate correlation coefficients between the self-assessed level of fatigue and all other factors evaluated by the DSS: stress, daytime sleepiness, sleep quality, mood on awakening, alertness on awakening, and sleep duration. Small circles on the plot represent the correlation values for each individual subject (n=18 per boxplot). Grey horizontal lines demarcate the thresholds for classifying weak (-/+), moderate (--/++), and strong (---/+++) levels of correlation (negative/positive, the symbols are shown on left sides of the boxplots next to the mean values, $p<0.05$).

The sleep quality indices taken from the PSD (sleep quality, mood on awakening, and alertness on awakening) show a moderate positive correlation between each other (standardized $\alpha_r = 0.754$ for the three items) and weak positive correlation with sleep duration (0.260±0.174, 0.316±0.164, and 0.395±0.214 respectively). All macro-averaged bivariate correlation values are summarized in Table II. Figure 3 shows distributions of the correlation coefficients for all pairs of the DSS indices that include the self-assessed level of fatigue across different subjects. The DASS-21 components in the group correlate as 0.773, 0.842, and 0.823 for the depression-stress, depression-anxiety, and anxiety-stress pairs, respectively.

TABLE III
MULTIPLE REGRESSION FIT FOR STANDARDIZED DAILY FATIGUE LEVEL AS THE OUTCOME VARIABLE

Factor	Value	Standard Error	t-value	p-value
Days from semester start	7.023×10^{-5}	20.817×10^{-5}	0.33	0.736
Stress	0.521	0.018	28.24	$< 0.001^{***}$
Sleepiness	0.211	0.019	11.06	$< 0.001^{***}$
Sleep quality	−0.059	0.020	−2.97	0.003^{**}
Mood	0.036	0.023	1.60	0.109
Alertness	−0.137	0.023	−5.90	$< 0.001^{***}$
Sleep duration	−0.057	0.019	−3.05	0.002^{**}

C. Multiple Regression Analysis

For inter-subject analysis, each of the following daily measures: fatigue, stress, daytime sleepiness, nighttime sleep duration, sleep quality, mood and alertness on awakening, were standardized by subtracting calculated over the entire study duration for each subject mean value and subsequent division by the standard deviation. The obtained z-scores were used for fitting a multiple regression model considering the self-assessed fatigue score as the output variable, with the other DSS components, as well as the number of days from the semester beginning, as independent factors. Predicting the fatigue level as based on the described set of independent factors results in r^2=0.486 and RMSE=0.709. The regression analysis indicates a significant positive slope in relation to the fatigue level for such factors as stress and daytime sleepiness, and negative slope for sleep quality, alertness of awakening and sleep duration. Table III summarizes the multiple regression fit results. While the time elapsed since the semester beginning does not affect the perceived fatigue level, the participants exhibited a trend for reporting lower fatigue

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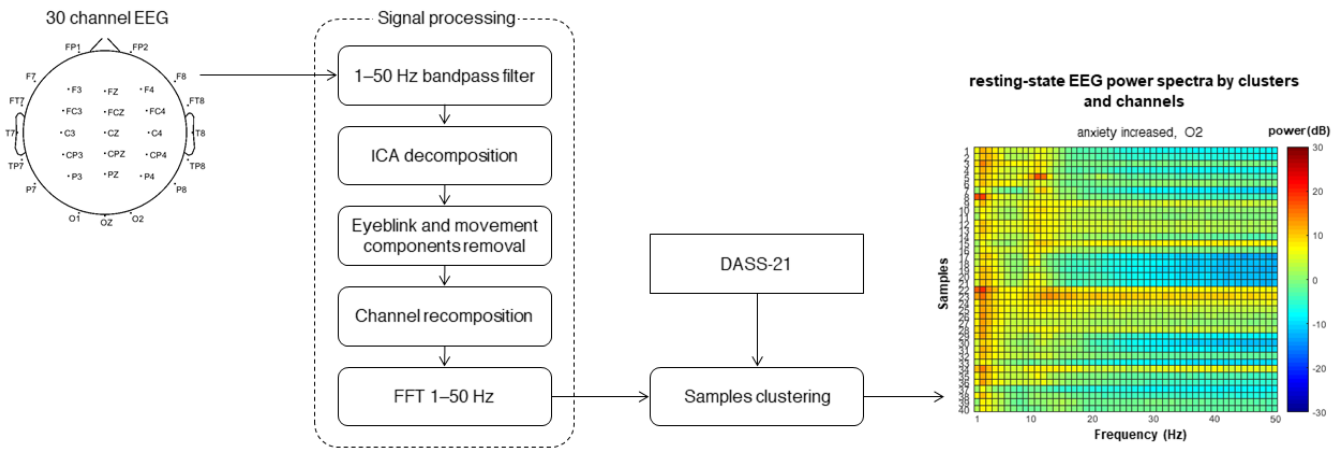

Fig. 4. The EEG data were collected from 30 electrodes placed according to the extended 10-20 system with average mastoid reference. The recorded signals were inspected for artifacts and dropped channels, bandpass filtered to remove slow signal drifts and power-line noise, and subjected to ICA decomposition to reject eyeblink and movement components. After the ICA cleaning, the samples were transformed into the power-frequency domain using FFT and grouped according to the levels of depression, anxiety, and stress using the DASS-21 scores.

scores on Saturdays (z-scored mean±SD equals -0.23 ± 0.97, $p<0.001$). Moreover, the reported stress level shows regular weekly oscillations with elevations on Mondays and Tuesdays (0.16 ± 0.95, $p=0.006$ and 0.13 ± 1.02, $p=0.030$ respectively) and decrease on Saturdays (-0.24 ± 0.90, $p<0.001$). The diagram with the distribution of the standardized stress scores by the days of the week, partial regression plots for each of the independent variables and combined multiple regression plots are included in supplementary materials.

D. Resting-State EEG and DASS-21

The EEG data collected in this study were inspected for signal quality and 93 records were selected for subsequent analysis. After the initial 1–50 Hz bandpass filtering for removing a slow baseline drift and power grid noise, the samples were subjected to independent components (IC) decomposition using the infomax ICA algorithm with optimizations [44], [45]. Eyeblink and movement components were manually removed from each record and the cleaned signals were reprojected to the channel representation using a standardized head model for the group study. The resulting EEG samples were converted into power spectra using FFT, with a step of 1 Hz in the 1-50 Hz range, and grouped according to the related DASS-21 scores.

Interpretation of the DASS test was performed using the original test methodology that sets up cut-off scores for severity ratings from 'normal' to 'extremely severe' for each scale based on the norms derived from a non-clinical sample predominantly composed of students [46]. To investigate the changes in the resting-state EEG power spectrum that accompanied an increase in the depression, anxiety, or stress levels, all DASS-21 samples were separated into 'normal' or 'increased' clusters, depending on the corresponding scores that were lower than or exceeded the upper threshold for the 'normal' level of each emotional condition. In total, $n_d=17$ samples were categorized to the state of increased depression, $n_a=40$ for increased anxiety, and $n_s=22$ for increased stress level. The EEG data processing and clustering sequence are illustrated as the flowchart in Figure 4.

The spectral analysis of the resting-state EEG data collected during this study reveals significant alterations in power spectral density associated with emotional state changes. Figure 5 illustrates differences in the spectral properties of the electrocortical oscillatory activity between the states of normal and increased levels for each of the three DASS-21 measures. Across subjects and sessions, the state of anxiety is characterized by an increase in the theta power detected over the entire head, and low alpha power at the prefrontal sites. An increase in the stress level corresponds to a positive response in the high delta, theta, and alpha oscillations, which is stronger over the prefrontal areas and weaker over the occipital areas. In addition, a significant suppression of the high-frequency spectra can be observed over the occipital region. The state of depression exhibits a strong positive shift in the resting-state EEG power over both temporal lobes, as well as an increase in the theta and alpha power over the frontal and prefrontal regions. The results of statistical tests for each EEG channel and frequency band are summarized in Figure 6.

The asymmetry scores for the prefrontal alpha band power were calculated as a relative difference of the power spectral density values averaged across the 8-13 Hz range, using a logarithmic power scale [47]. No significant differences in the prefrontal asymmetry were observed when comparing the states of normal and increased levels of depression, anxiety, or stress. Negative group averaged prefrontal asymmetry scores are associated with all conditions, except the state of increased depression, which denotes left hemispheric dominance in alpha spectral power: 95% confidence intervals of the mean values are -0.396-0.066 and -0.059-0.073 for no depression and depression, -0.215-0.063 and -0.312-0.097 for no anxiety and anxiety, and -0.161-0.095 and -0.359-0.082 for no stress and stress conditions.

IV. Discussion

Students often experience stressful events due to educational and research tasks, such as homework, exams, or course projects. Graduate students commonly take fewer course credits as compared to undergraduate students and follow a more

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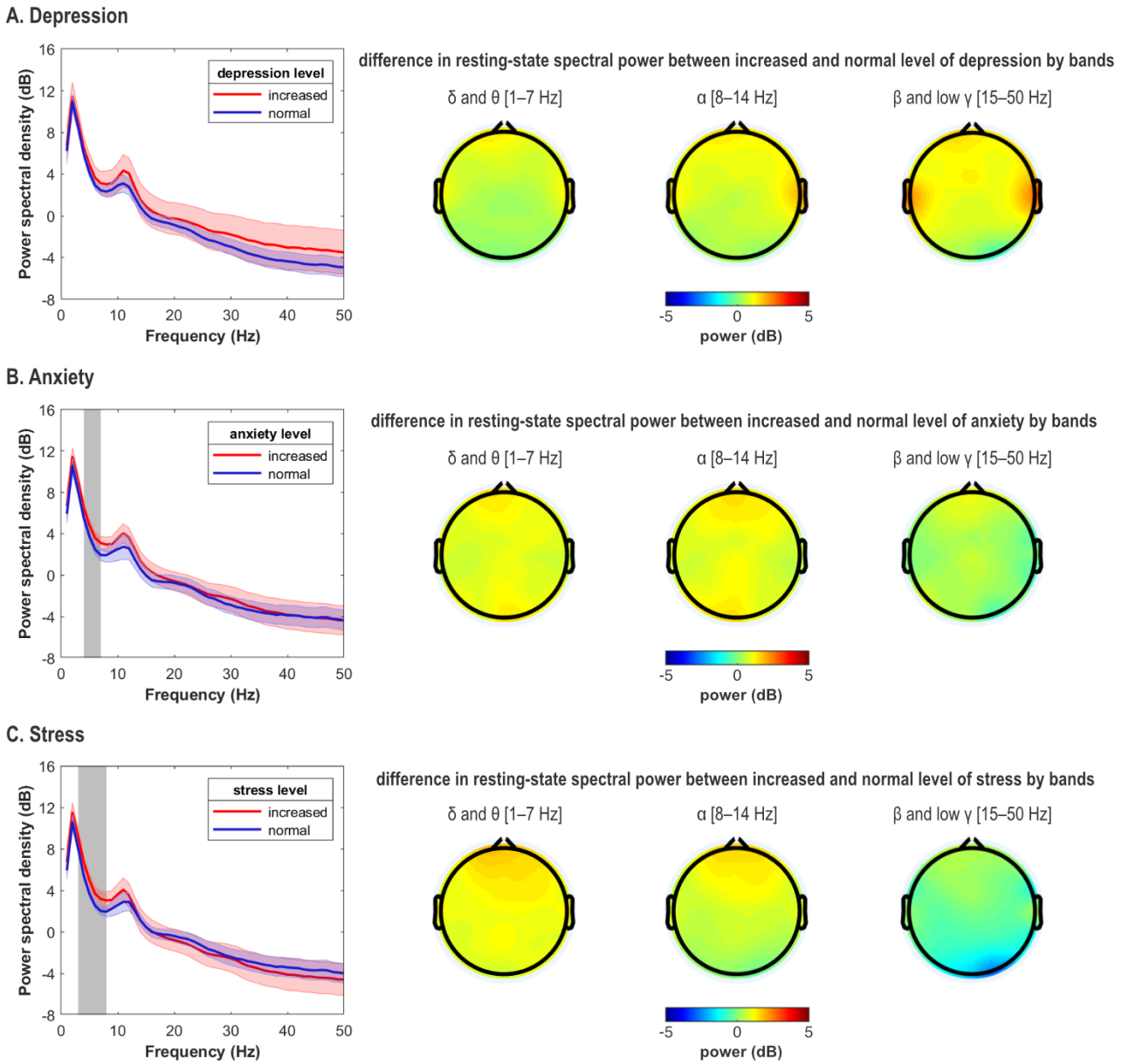

Fig. 5. Differences in resting-state EEG spectral characteristics between the conditions of increased and normal according to the DASS-21 levels of (A) depression, (B) anxiety and (C) stress. The power spectra plots illustrate averaged for all sessions within each cluster integral EEG power spectrum after non-brain related components rejection and respective 95% confidence intervals of the mean. The grayed areas outline the frequencies with a significant difference in the PSD values between the two conditions ($p<0.05$ with FDR). The topoplots illustrate the difference in averaged resting-state EEG spectral power in different frequency bands as subtracting the normal from the increased level for each emotional state.

flexible schedule in the coursework arrangement. This may result in the absence of a regular waking time every morning, which is typically determined by the beginning of the work or school day, as exemplified by other social groups, such as white-collar workers, and undergraduate or high-school students. The original Pittsburgh Sleep Diary study reported a bimodal distribution of sleep durations in a group of healthy middle-aged subjects, which was caused by the presence or absence of work commitments on workdays and weekends respectively [35]. This phenomenon of noticeably different sleep durations between weekdays and weekends has also recently been demonstrated in a group of all-girls secondary school students in Singapore [48]. However, no such pattern was observed in the present study, indicating that such bimodal sleep schedules are not intrinsic for graduate students with flexible schedules. A survey-based study of sleep behavior in an American high school reports 7.3±1.2 hours of sleep duration in the baseline group, with no gender differences [49], which is close to the findings from the longitudinal data collected in this experiment.

The concept of fatigue is very general and widely referenced in everyday life. While short-term fatigue can be referred to as a deterioration in task-directed performance [50], prolonged or general fatigue is usually being considered a multi-factor syndrome related to a lack of energy, sleepiness, and health and emotional problems, which leads to an inability

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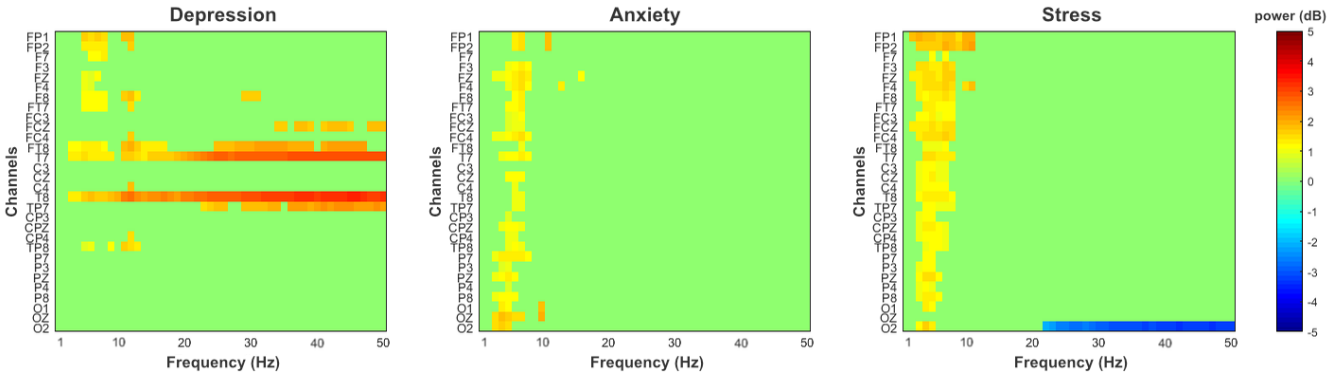

Fig. 6. Group-averaged differences in the resting-state EEG power spectral density between the increased and normal levels of depression, anxiety and stress for all channels as per extended international 10-20 system of electrodes placement, and frequencies in 1–50 Hz range with a step of 1 Hz. The difference is calculated by subtraction of the normal from increased group average PSD values for each cell. The cells that do not pass the $p<0.05$ with FDR statistical threshold are set to zero on the plot for visual clarity.

to maintain daily routines [51], [52]. This study reveals a strong positive correlation between self-evaluated measures of fatigue and stress, for all subjects. At the same time, while most of the participants reported associations among fatigue, daytime sleepiness, sleep duration, and mood on awaking, not all subjects followed the same trend. As was demonstrated in the general population studies, stress and depression are closely related to emotional instability and fatigue, and may act as independent risk factors for each other [32], [53]. This indicates that sleep duration and quality may have a significant, but not decisive influence on prolonged fatigue and subjective well-being, and that it is highly important to take emotional state factors into consideration in sleep and fatigue studies.

This study did not observe a significant difference between male and female subjects in the self-assessed levels of fatigue, stress, sleep quality, mood, alertness, and sleep duration. Similar conclusions were made in studies of sleep behavior in elementary and high school students [54], [55], nor were gender differences in sleep duration observed in meta-analyses of sleep behavior studies in people 58 years of age and older [56]. However, several studies based on the middle-age population reported such gender-specific differences [57], in particular, a higher prevalence of excessive daytime sleepiness in females [58], [59]. This leads to an assumption that differences in sleep behavior may be driven not only by directly gender-related physiological factors, but also by cultural, social, and family circumstances, such as childcare or employment, which are seldom specified in research reports. While this study's design was incapable of resolving this hypothesis, it should be verified for future research.

Longitudinal survey-based research has clear advantages over cross-sectional data collection, which is based on a snapshot of participants daily lives. Such an approach makes it possible to monitor variations in the investigated parameters over time in either single or multiple subjects, take into account factors that are often ignored otherwise, and adjust for spontaneous or short-term fluctuations. As it has been demonstrated in this study, the measures of subjective emotional well-being may fluctuate over the course of the week, causing a bias in the one subject, one report approach if the data are collected on different days or unequally distributed. Different association strengths between sleep quality and mental state indices in different subjects can be detected only if multiple reports are collected over time from each participant. In general, the proposed method can be easily adjusted for particular experimental needs and implemented as a smartphone application or web-based online system, making it easy to report, collect, and process the data, and as a result, such a method is more preferable over single report surveys. The presence of a progressive monetary reward for participation in the experiment and clear user interface, which simplifies the process of report submissions, are among the methods that are believed to increase the response rate [60]. None of the participants dropped out of the data collection, and the majority of them showed persistent daily report submission rates higher than 70% [61]. However, as this study lacked a control group that received no fee for the submitted reports, it is unable to assess a direct comparison between the reward and no reward approaches.

The results of EEG data analysis revealed significant alterations in the resting-state spectrum that accompanied the changes in the emotional state of the participants. These findings are consistent with other recent brain imaging studies of anxiety and depression, which particularly reported the presence of related EEG biomarkers over the frontal and temporal regions [62]. However, a direct comparison of neuroimaging findings across different experiments remains a challenge, primarily because of methodological differences in determining the control and experimental samples. A common approach is to compare two separate groups of healthy patients and ones diagnosed with depression and/or anxiety disorders, this can lead to inter-subject variability issues and discrepancies in the reported results [63]. As for short-term tasks, performed under controlled laboratory conditions to induce the feeling of stress, the extent to which the experimental task is experienced as stressful depends on how seriously and personally important the participant treats it. Such constraints mean that the laboratory stress-inducing tasks may not faithfully represent the stressors that are encountered in real life [64]. The present study implements longitudinal monitoring of the emotional state factors in a healthy group that is naturally influenced by realistic, educational stressors, which is an important step in

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transferring neuroimaging research towards real-world settings and directly generalizable results.

While variations in the resting-state EEG were widely suggested as appropriate markers for the emotional state prediction [65], [66], a meta-analytic review of the associations of anxiety, depression, and resting frontal EEG asymmetry [67] reported contradictory results among different studies, including in the null findings. This leads to an assumption that the presence or absence of the described effects may depend on multiple factors, ranging from differences in emotional state evaluation and brain imaging methods to participant demographics. This was partially confirmed in separate studies that linked the relationship of frontal asymmetries' implications to risk for anxiety and depression with the sex and age of the subjects [68], reference montages and signal transformations [69], and the influence of certain short-term factors that affect cognitive state, such as the amount of time since a participant had woken up at the time of data recording [70]. As a summary, reporting the neural correlates of different emotional states under various experimental conditions is beneficial for reconciling the inconsistencies of previous findings in future meta-analyses.

V. Conclusion

This study implemented longitudinal, multi-component data collection in a group of graduate school students throughout an academic semester. The dataset included self-evaluated on a daily basis emotional state and sleep quality scores, as well as resting-state EEG data and DASS-21 measures. The results demonstrated a positive association of subjective fatigue levels with stress and daytime sleepiness, and a negative association with sleep quality and duration, and mood and alertness on awakening. A higher variance in the sleep duration in comparison with such social groups as high school students or white-collar workers can be attributed to the lack of regular working hours due to the elective course system used in the university. The increase in the levels of depression, anxiety, and stress in the experimental group is accompanied by significant alterations in the resting-state EEG spectral power, indicating the presence of electrophysiological markers that can be used to detect the related states of emotional arousal. However, no significant difference in the prefrontal alpha power asymmetry was observed under the given experimental settings between the normal, calm states and any of the depression, stress, or anxiety conditions. Considering a high response rate in the daily sampling report submissions, the proposed approach could be implemented in further larger scale studies.

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